

Type Through the Bible (C++ Edition)

A Typing Game of Biblical Proportions

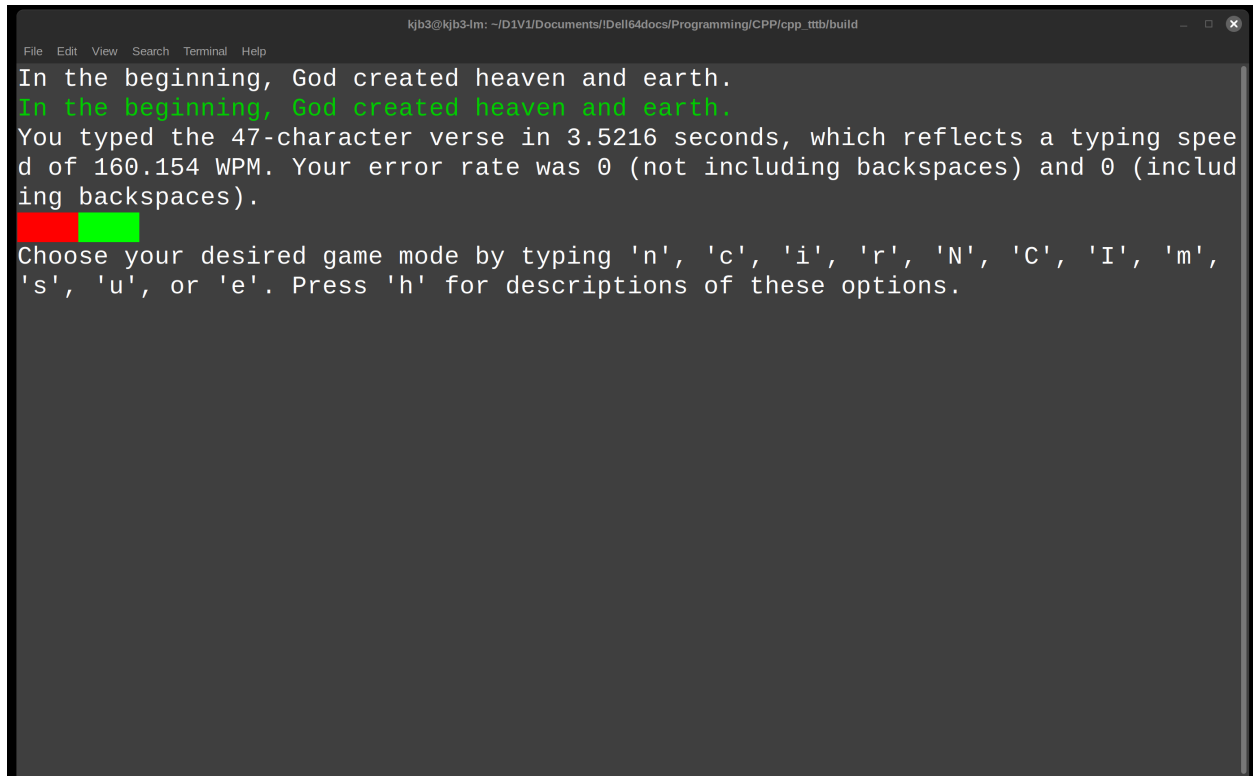
By Ken Burchfiel

Released under the MIT License

[Link to GitHub repository](#)

[Link to game download page on Itch.io](#)

[Link to gameplay demo video](#)



```
kjb3@kjb3-lm: ~/Documents/IDell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
In the beginning, God created heaven and earth.
In the beginning, God created heaven and earth.
You typed the 47-character verse in 3.5216 seconds, which reflects a typing speed of 160.154 WPM. Your error rate was 0 (not including backspaces) and 0 (including backspaces).
Choose your desired game mode by typing 'n', 'c', 'i', 'r', 'N', 'C', 'I', 'm', 's', 'u', or 'e'. Press 'h' for descriptions of these options.
```

Latest versions:

C++ code: 1.02 Complementary Python code: 1.01

Introduction

Type Through the Bible (TTTB) allows you to practice your keyboarding skills by typing the Bible verse by verse. It contains both single-player and multiplayer modes; in addition, it offers (via a complementary Python script) a wide variety of interactive visualizations.

Download instructions

Linux, Windows, and Apple-silicon OSX binaries of TTTB are available [at this page](#) on itch.io . The game is free to download, but donations (while not expected) are greatly appreciated.

Make sure to download both the zipped folder for your operating system *and* the corresponding README (e.g. the file you're reading right now). Once you've unzipped the folder on your local computer, you can move it to a location of your choice—or simply keep it within the downloads folder. Nothing needs to be installed in order for you to begin playing the game.

(If multiple people will be playing TTTB's single-player mode on your computer, I recommend creating a separate copy of the game for each of them. This will make it much easier for each player to keep track of his or her progress.)

Because TTTB is released under the MIT license, you are free to modify its underlying C++ and Python code, then share your revisions with others (even under a proprietary license). In this case, you'll want to download and compile the source code rather than use these underlying binaries. See 'Compilation instructions' below for more details.

Mac/OSX users: Make sure to carefully review 'Starting the game' section below for important details on successfully launching TTTB.

Updating your default configuration settings

Once you've downloaded the game, you may want to update your default configuration settings before you begin. That way, certain default values (like your name) will get stored within every test; you'll also have the option to update these settings within single-player games. You can add these default options within the second row (e.g. the row right below the headers) of the `game_config.csv` file.

Here's what my default settings look like: (I use `Tag_1` to store the keyboard I'm using for a test and `Tag_2` to store my location. You can choose not to enter any default tags, but I'd recommend at least storing your name within the 'Player option'.)

	A	B	C	D	E
1	Player	Tag_1	Tag_2	Tag_3	Notes
2	KJB3	Cherry_Red	Home		
3					

TTTB's folder directory

Here is how TTTB's folders and files are organized:

`cpp_tttb/` [the root folder for the game; your exact name may differ]

—**build/**

—`_internal/` [contains Pyinstaller-related files; these are crucial to include in order for the corresponding Python executable to work correctly, but you shouldn't need to modify them. This folder isn't part of the OSX release, as its contents are compressed within the `tttb_py_complement` executable.]

—`tttb` [`tttb.exe` on Windows; the main TTTB executable that you'll launch in order to start the game]

—`tttb_py_complement` [`tttb_py_complement` on Windows; the Python executable that TTTB uses to (1) create single-player and multiplayer visualizations and (2) combine various multiplayer files into single files. This executable can be called either within TTTB or as a standalone executable.]

—**Files/**

—`MP_Test_Result_Files_To_Combine/`

—`MP_Word_Result_Files_To_Combine/`

—`Multiplayer/` [multiplayer results will be stored here.]

—`CPDB_for_TTTB.csv`

—`game_config.csv` [This file stores default configuration options, such as your name and various gameplay tags.]

——test_results.csv

——word_results.csv

——**Visualizations/**

——Multiplayer/ [Multiplayer visualizations will get stored here.]

——Single_Player/ [Single-player visualizations will get stored here.]

Gameplay instructions

Starting the game

To launch TTTB, you'll first want to open up your command prompt or terminal, then navigate to the program's **build** folder (where the game's 'tttb'/'tttb.exe' executable file is located). On my Linux computer, I can accomplish this via the following steps:

1. Press Ctrl + Alt + T to launch my terminal
2. Enter `cd '/home/kjb3/D1V1/Documents/!Dell64docs/Programming/CPP/cpp_tttb/build'` to navigate to the build folder. (This folder path will of course look different on your end—but it should end in `build/`, not `cpp_tttb` or whatever your root TTTB folder is called.)

The Mac steps look very similar; you'll just need to search for your Terminal app if it's not already present within your taskbar.

On Windows, I would instead press the Windows button, then enter `cmd` to bring up the command prompt. I would then enter `cd` followed by the path to my build folder (perhaps encased in double quotes).

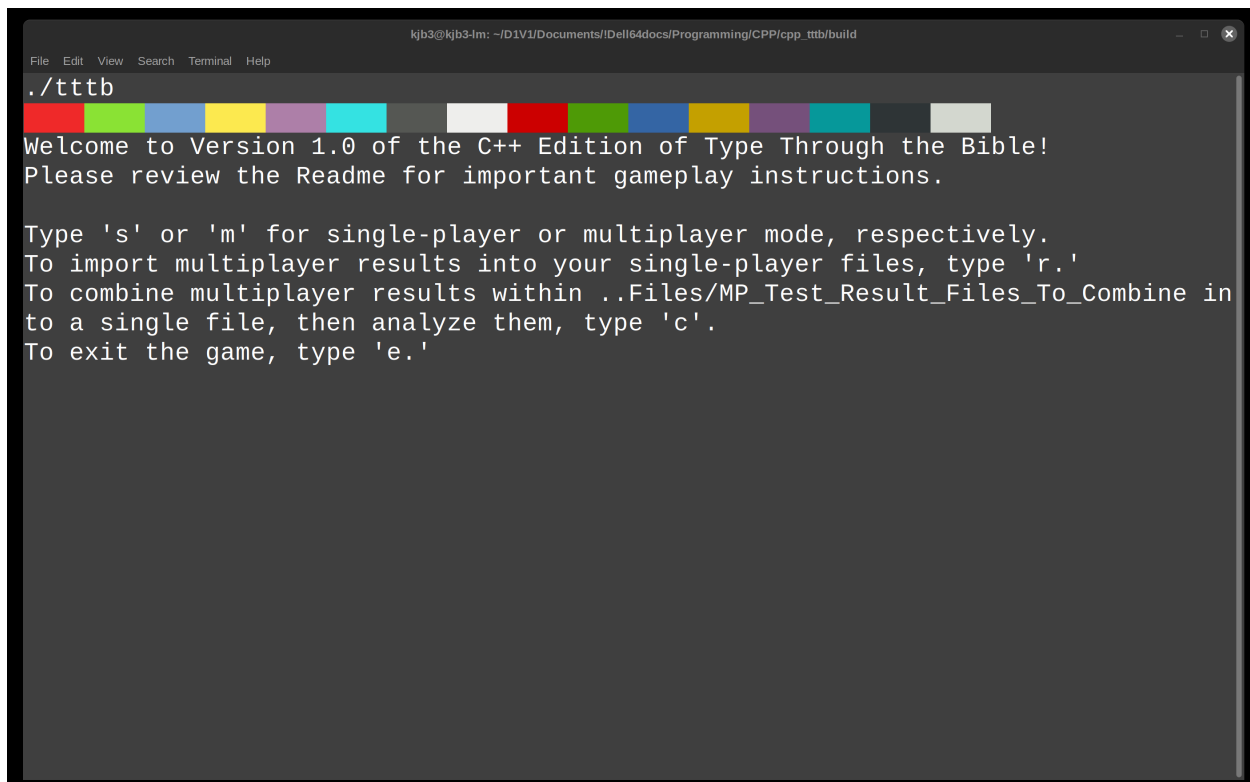
Once you've navigated to this folder, you can launch TTTB by entering the following command (on Linux or OSX):

```
./tttb
```

On Windows, you would instead enter:

```
tttb.exe
```

Once you've completed these steps, you should see the following welcome screen:

A screenshot of a terminal window titled "kjb3@kjb3-lm: ~/D1V1/Documents/Idell64docs/Programming/CPP/cpp_ttb/build". The terminal shows the command `./tttb` being executed. Below the command is a decorative bar with a series of colored squares. The text in the terminal reads: "Welcome to Version 1.0 of the C++ Edition of Type Through the Bible! Please review the Readme for important gameplay instructions. Type 's' or 'm' for single-player or multiplayer mode, respectively. To import multiplayer results into your single-player files, type 'r.' To combine multiplayer results within ../Files/MP_Test_Result_Files_To_Combine in to a single file, then analyze them, type 'c'. To exit the game, type 'e.'".

```
kjb3@kjb3-lm: ~/D1V1/Documents/Idell64docs/Programming/CPP/cpp_ttb/build
./tttb
Welcome to Version 1.0 of the C++ Edition of Type Through the Bible!
Please review the Readme for important gameplay instructions.

Type 's' or 'm' for single-player or multiplayer mode, respectively.
To import multiplayer results into your single-player files, type 'r.'
To combine multiplayer results within ../Files/MP_Test_Result_Files_To_Combine in
to a single file, then analyze them, type 'c'.
To exit the game, type 'e.'
```

However, things are a bit trickier on OSX. When you launch the game *for the first time*, you'll see a box pop up with the message "tttb Not Opened." Here's how you can resolve this issue:

- Click on the question mark on the top right of this box to open another window with the title "Apple can't check app . . .".
- Select 'Open Privacy & Security settings for me' within this window.
- Close the "Apple can't check app" window; hit 'Done' on the original "tttb Not Opened" box; and then, within the Privacy & Security Settings window; select the 'Allow Anyway' option next to the "'tttb' was blocked" message.
- Within your terminal, re-enter the `./tttb` command. (You should be able to get back to it by pressing the up arrow.) You'll now see an "Open 'tttb'?" window come up.
- Click "Open anyway", then enter your account password.
- You're almost done! You'll likely see a "zsh: segmentation fault" message appear in your terminal. If this occurs, simply enter `./tttb` in your terminal one more time, and you should finally be able to enter the game.

Thankfully, during subsequent gameplay sessions, you should *not* need to repeat this process. *However*, at the end of your first session, you'll need to repeat the same steps for the `tttb_py_complement` executable that gets called to run Python-based data visualization scripts. (After you've completed these steps once, you shouldn't need to perform them again.)

An alternative to all these steps is to download the GitHub clone of TTTB and compile both the C++ and Python code yourself. (See Compilation instructions section below for more details.)

[Note: Windows may also prompt you for your approval before allowing TTTB, or the `tttb_py_complement.exe` executable, to run. You should only need to do this a few times, but if you keep getting prompted for different Python libraries, let me know by creating a new issue. I can then replace the existing Windows Pyinstaller setup with a single-file setup that I use for the OSX release.]

Single-player gameplay

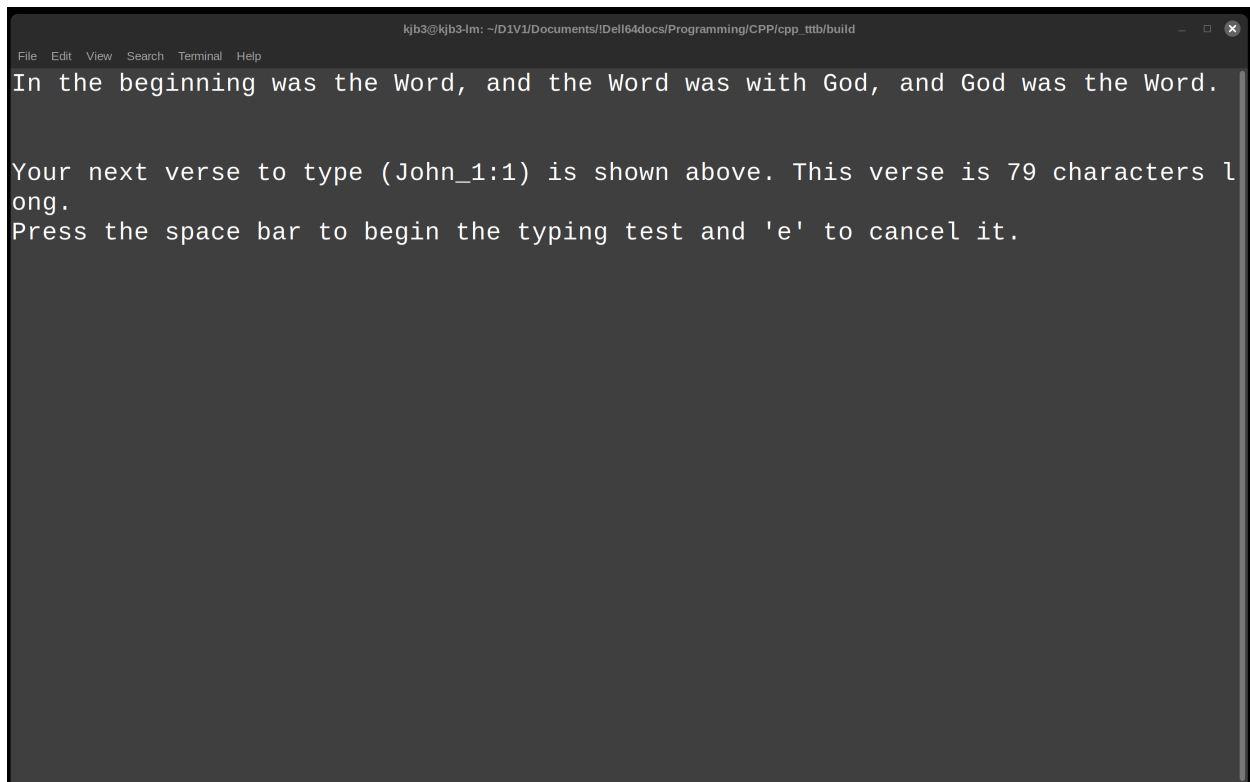
To launch single-player games, press ‘s’ within the welcome screen. You’ll then be presented with a number of different single-player modes from which to choose:

- **n**: Type the next untyped verse, then return to the gameplay menu. (This, along with ‘N’, is a good starting option.)
- **c**: Type the next verse, then return to the gameplay menu. (The main difference between ‘c’ and ‘n’ is that the former won’t skip over verses that you’ve already typed at least once.)
- **i**: Type a specific verse ID, then return to the gameplay menu.
- **r**: Repeat the verse you just typed, then return to the gameplay menu.
- **N**: Automatically get directed to the next untyped verse. (This option eliminates the need to press ‘n’ after each test if you wish to stay in this mode.)
- **C**: Automatically get directed to the next verse.
- **I**: Select a verse from which to begin, then automatically get directed to the verses that follow it.
- **m**: Enter ‘untyped marathon mode,’ in which you will immediately start a test for the next untyped verse after finishing the current test. (This option, along with ‘S’, can get pretty exhausting—so consider trying it out only for short periods. Also note that, in the stats that appear under your entry within all marathon modes, ‘E&B’ stands for ‘error and backspace rate’. See ‘Understanding error rates’ section below for more information.)
- **s**: Enter ‘sequential marathon mode,’ in which you will immediately start (1) the verse following the one you just typed or (2) (when starting this mode) a verse of your choice.
- **U**: Update game configuration settings.
- **C**: Exit this session and save your progress.

Note: You can exit out of any test, even in marathon mode, by pressing Ctrl+C (not to be confused with Command+C). **Do not** simply exit out of the terminal, as your progress then won’t be saved.

Running typing tests

Within each game mode, you’ll be presented with one or more tests. Each of these tests will consist of a single Bible verse that you’ll need to type. Unless you’re in one of the two ‘marathon’ modes, you’ll have the chance to review the verse before you begin typing it:

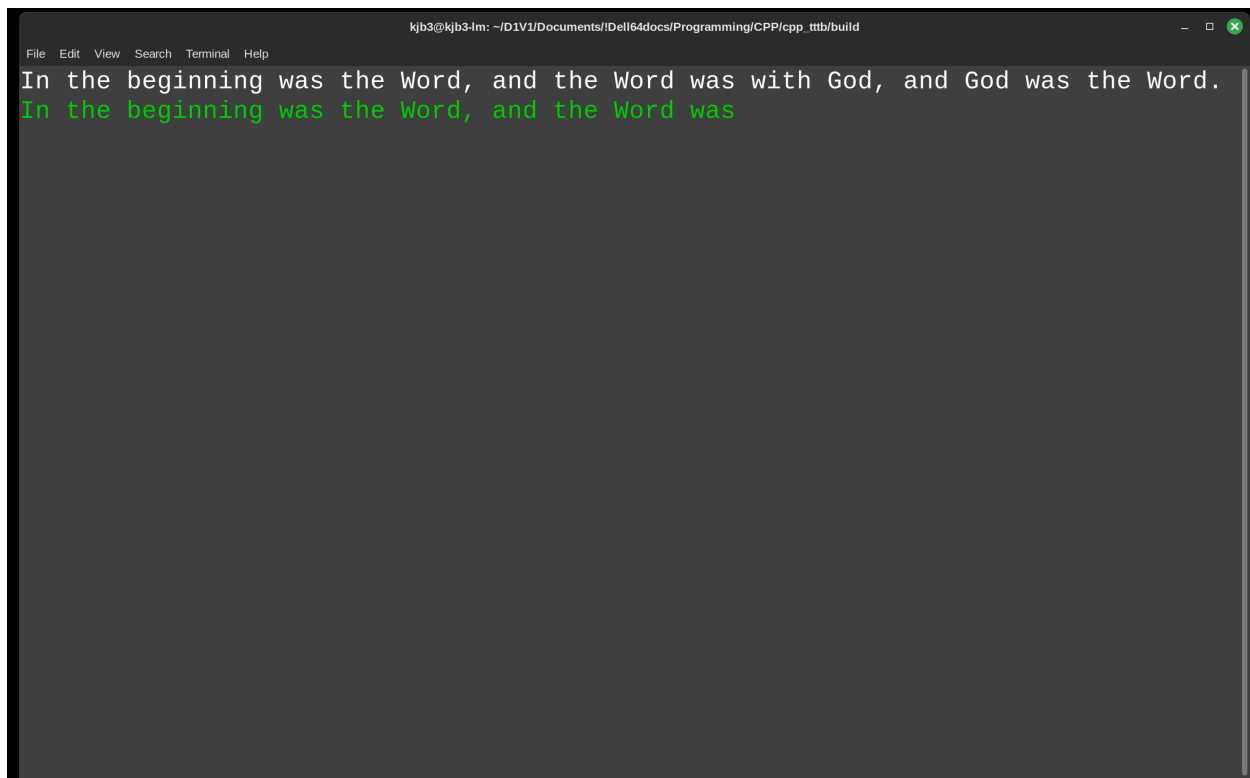
A screenshot of a terminal window with a dark background. The window title bar shows the user 'kjb3' and the path '~/.D1V1/Documents/IDell64docs/Programming/CPP/cpp_ttb/build'. The terminal has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The text in the terminal reads: 'In the beginning was the Word, and the Word was with God, and God was the Word.' followed by 'Your next verse to type (John_1:1) is shown above. This verse is 79 characters long.' and 'Press the space bar to begin the typing test and 'e' to cancel it.'

```
kjb3@kjb3-lm: ~/.D1V1/Documents/IDell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
In the beginning was the Word, and the Word was with God, and God was the Word.

Your next verse to type (John_1:1) is shown above. This verse is 79 characters long.
Press the space bar to begin the typing test and 'e' to cancel it.
```

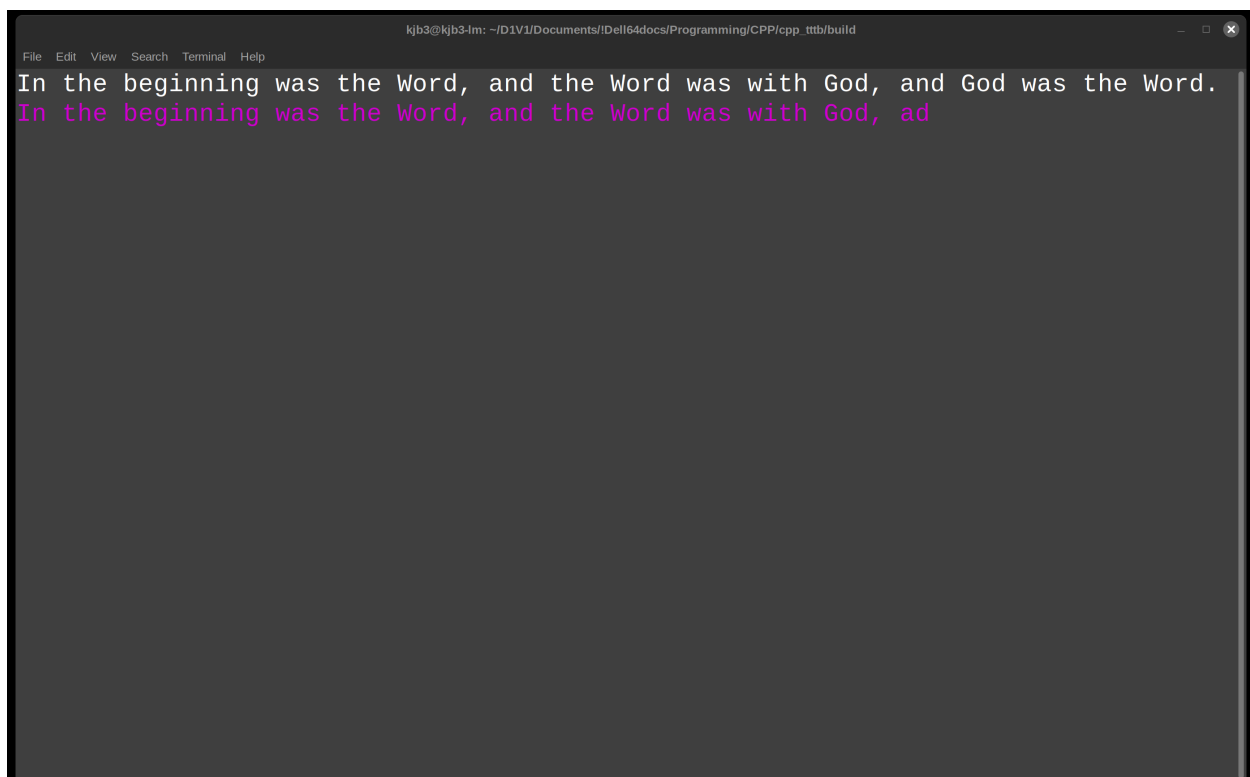
Once you press the space bar, the test's timing clock will start and you'll be able to begin typing. This timing clock will end automatically once you have typed the verse correctly. (In other words, you won't need to press 'Enter' after the test is over.)

You'll get color-coded feedback as you progress through the test. As long as you've typed the verse correctly, your response will be highlighted in green:

A screenshot of a terminal window with a dark background. The window title is 'kjb3@kjb3-lm: ~/D1V1/Documents/Dell64docs/Programming/CPP/cpp_ttb/build'. The menu bar includes 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The text 'In the beginning was the Word, and the Word was with God, and God was the Word.' is displayed in white. Below it, the text 'In the beginning was the Word, and the Word was' is displayed in green. The terminal has a scrollbar on the right side.

```
kjb3@kjb3-lm: ~/D1V1/Documents/Dell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
In the beginning was the Word, and the Word was with God, and God was the Word.
In the beginning was the Word, and the Word was
```

However, if you make a typo (even as minor as an extra space), your response will turn magenta in color:

A screenshot of a terminal window with a dark background. The window title is 'kjb3@kjb3-lm: ~/D1V1/Documents/Dell64docs/Programming/CPP/cpp_ttb/build'. The menu bar includes 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The text 'In the beginning was the Word, and the Word was with God, and God was the Word.' is displayed in white. Below it, the text 'In the beginning was the Word, and the Word was with God, ad' is displayed in magenta. The terminal has a scrollbar on the right side.

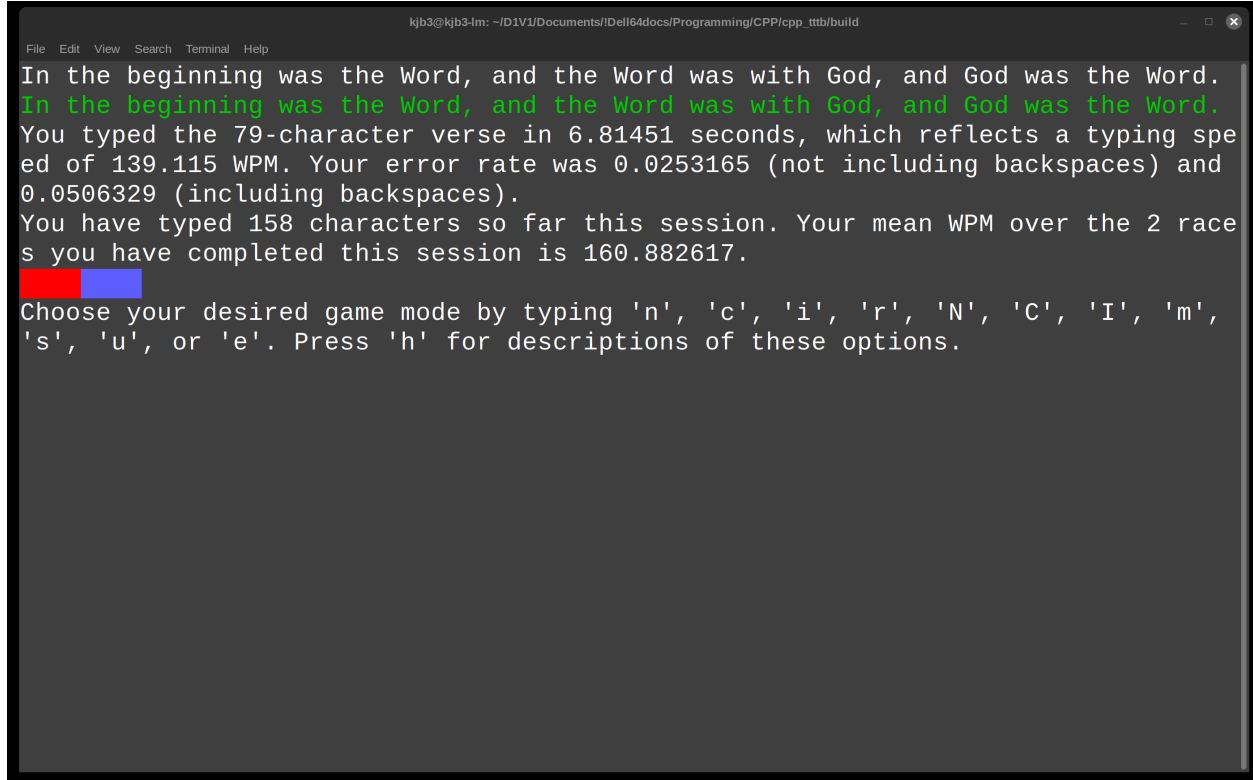
```
kjb3@kjb3-lm: ~/D1V1/Documents/Dell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
In the beginning was the Word, and the Word was with God, and God was the Word.
In the beginning was the Word, and the Word was with God, ad
```

Once you correct the typo (e.g. by hitting backspace), the text will turn green again.

Important: To clear out an entire mistyped word, press Alt + Backspace on Linux and Windows or Fn

+ Delete on OSX. (I would have liked to make Ctrl + Backspace an option for Linux and Windows, and Command + Delete an option for OSX, but I had trouble implementing these options using the cpp-terminal library; it's possible that they might be added in in a future update.)

After you finish the verse, you'll get feedback on how long it took you type the text; your WPM (defined as $((\text{characters in the verse}) / (\text{time, in seconds, you needed to complete the test})) * 60 / 5$; and your error rates.



```
kjb3@kjb3-lm: ~/D1V1/Documents/Idell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
In the beginning was the Word, and the Word was with God, and God was the Word.
In the beginning was the Word, and the Word was with God, and God was the Word.
You typed the 79-character verse in 6.81451 seconds, which reflects a typing speed of 139.115 WPM. Your error rate was 0.0253165 (not including backspaces) and 0.0506329 (including backspaces).
You have typed 158 characters so far this session. Your mean WPM over the 2 races you have completed this session is 160.882617.
Choose your desired game mode by typing 'n', 'c', 'i', 'r', 'N', 'C', 'I', 'm', 's', 'u', or 'e'. Press 'h' for descriptions of these options.
```

Understanding error rates

TTTB keeps track of two error rates for each verse: one that counts backspaces (which, while not necessarily errors, will still affect your final typing speed), and one that doesn't. The denominator for each rate is the number of characters in the verse. Thus, an error rate of 0.05 means that you had 5 erroneous keypresses for every 100 characters.

Once you're ready to quit out of the game, press 'e'. Your results will then be saved to .csv files.


```
kjb3@kjb3-lm: ~/D1V1/Documents/IDell64docs/Programming/CPP/cpp_tttb/build
File Edit View Search Terminal Help
In the beginning was the Word, and the Word was with God, and God was the Word.
In the beginning was the Word, and the Word was with God, and God was the Word.
You typed the 79-character verse in 6.81451 seconds, which reflects a typing speed of 139.115 WPM. Your error rate was 0.0253165 (not including backspaces) and 0.0506329 (including backspaces).
You have typed 158 characters so far this session. Your mean WPM over the 2 races you have completed this session is 160.882617.
Choose your desired game mode by typing 'n', 'c', 'i', 'r', 'N', 'C', 'I', 'm', 's', 'u', or 'e'. Press 'h' for descriptions of these options.
Exiting game and saving progress.
Exported 35839 Bible verses in 0.244292 seconds.
Exported 2 test result(s) in 7.3568e-05 seconds.
Exported 34 word results in 0.000142273 seconds.

Press 'y' to call a Python script that updates single-player stats; type 'n' to skip this process. (This may require some setup on your part; see Readme for more details. You can also run that script separately if you prefer.
```

You'll then have the opportunity to call a Python script that will create interactive visualizations of these results. This shouldn't take terribly long, but if you're in a rush and/or have completed massive numbers of tests, you may not want to call it right now. (You can always call it later; since the script retrieves its data from the .csv files in the Files/ folder, which *always* get updated following a successful exit from single-player mode, declining to visualize your results at this time won't cause any data to get lost.)

```
kjb3@kjb3-lm: ~/D1V1/Documents/Idell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
Exported 35839 Bible verses in 0.244292 seconds.
Exported 2 test result(s) in 7.3568e-05 seconds.
Exported 34 word results in 0.000142273 seconds.

Press 'y' to call a Python script that updates single-player stats; type 'n' to
skip this process. (This may require some setup on your part; see Readme for mor
e details. You can also run that script separately if you prefer.
Calling Python script. (This can take a little while.)
Starting Python script. It may take a little while to run; please be patient.
Analyzing single-player results.
Analyzing WPM data.
Analyzing endurance data.
Analyzing accuracy data.
Analyzing word-level data.
Analyzing overall progress data.
Finished calculating and visualizing single-player stats in 1.59 seconds.
Quitting single-player gameplay session.

Type 's' or 'm' for single-player or multiplayer mode, respectively.
To import multiplayer results into your single-player files, type 'r.'
To combine multiplayer results within ../Files/MP_Test_Result_Files_To_Combine in
to a single file, then analyze them, type 'c'.
To exit the game, type 'e.'
```

Autosave files Every 10 races, TTTB will automatically save test-level and word-level results, along with an updated copy of your CPDB_for_TTTB.csv file, to your Files/ folder. If the game happens to crash, *and* you have completed at least 10 races, you *may* want to:

1. Manually add the data in autosaved_test_results.csv and autosaved_word_results.csv to your main word_results.csv and test_results.csv files. (This will involve opening up both sets of files, then copying the rows from the autosave file into the bottom of the main files.)
2. Replace CPDB_for_TTTB.csv with autosaved_CPDB_for_TTTB.csv.

If, on the other hand, you haven't completed at least 10 races, you *won't* want to copy in these autosave files, as they will reflect data from previous sessions.

Multiplayer gameplay

Type Through the Bible also supports multiplayer gameplay. There are two methods of performing multiplayer gameplay: single-computer and multi-computer.

Single-computer multiplayer gameplay

In this method, players will take turns typing on the same computer. This makes for a simpler setup, but is less efficient (at least for larger numbers of players) than the multi-computer setup that I'll discuss shortly.

To begin this mode, press 'm' within TTTB's start menu to launch a multiplayer game. You'll be prompted to specify (1) the names of each player; (2) the number of rounds, and tests within rounds, you'd like to play; and (3) the verse ID at which you'd like to begin the game. (There's also an option to type randomly-selected verses. Regardless of whether you wish to type consecutive or randomly-chosen verses, each player will type the exact same sequence of passages.)

```
kjb3@kjb3-lm: ~/D1V1/Documents/IDell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help

Welcome to Type Through the Bible's multiplayer mode! First, enter the names of
all players one by one; make sure to press Enter after each name. (Player names
should not include whitespace.) Once all names have been entered, type 'c' and
hit Enter.

Ken
Added Ken to player list. You may now enter the next player's name.

Keith
Added Keith to player list. You may now enter the next player's name.

Kevin
Added Kevin to player list. You may now enter the next player's name.

c
Here are the 3 players who will be joining this game:
■ Ken
■ Keith
■ Kevin
```

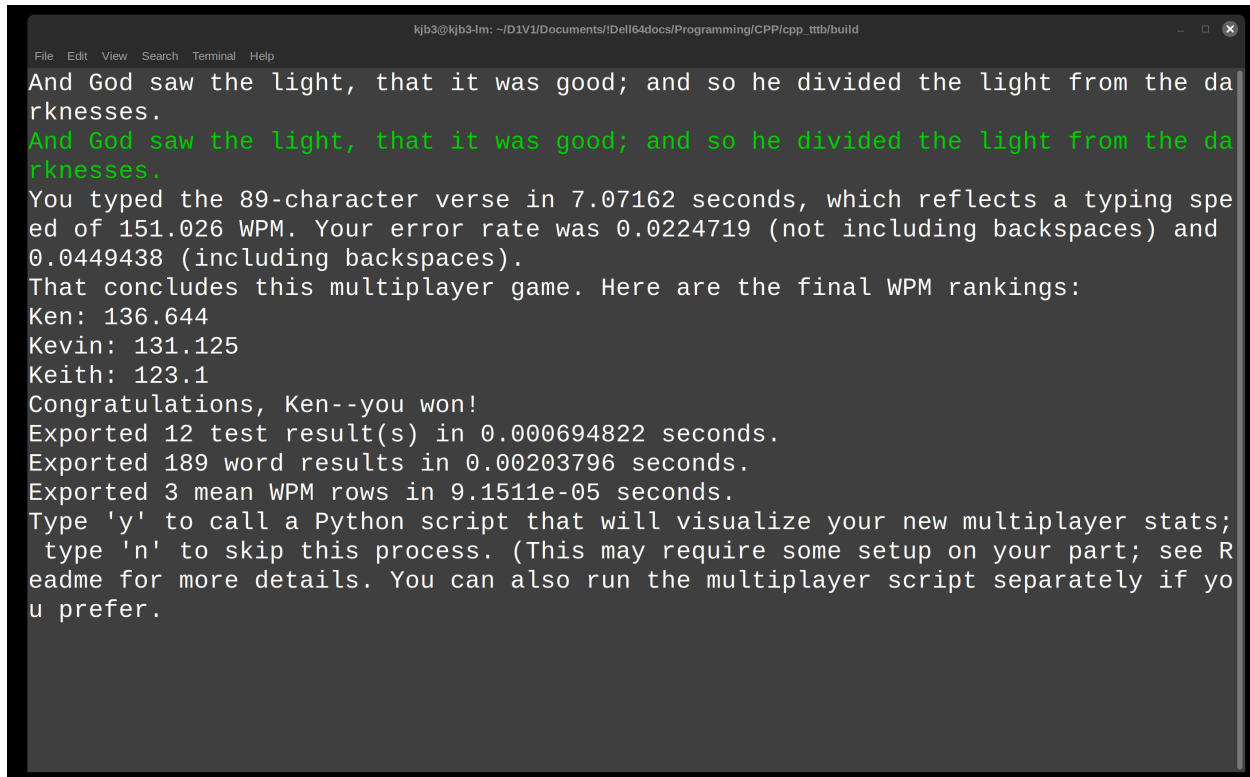
Players will then take turns typing the series of Bible verses that you specified during setup. Text notifications will inform you who is typing; make sure to pay careful attention to them so that you don't accidentally type someone else's race (which I did once!).

```
kjb3@kjb3-lm: ~/D1V1/Documents/IDell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help

But the earth was empty and unoccupied, and darknesses were over the face of the
abyss; and so the Spirit of God was brought over the waters.
But the earth was empty and unoccupied, and darknesses were over the face of the
abyss; and so the Spirit of God was brought over the waters.
You typed the 141-character verse in 12.3117 seconds, which reflects a typing sp
eed of 137.43 WPM. Your error rate was 0.0212766 (not including backspaces) and
0.0425532 (including backspaces).
That concludes the current player's turn for the current round. Here are the WPM
rankings thus far:
Kevin: 132.791
Ken: 132.545
Keith: 120.162

Here are the details for the following test:
Round: ■ 2
Player: ■ Ken
Test within round: ■ 1
Press 'c' to continue. (The test won't start just yet.)
```

Once the game ends, you'll get to see everyone's mean WPM; the player with the highest average WPM is the winner. The game will also (1) save a series of multiplayer results files to Files/Multiplayer and (2) ask whether you wish to create visualizations of these files.



```
kjb3@kjb3-lm: ~/D1V1/Documents/IDell64docs/Programming/CPP/cpp_tttb/build
File Edit View Search Terminal Help
And God saw the light, that it was good; and so he divided the light from the da
rkneses.
And God saw the light, that it was good; and so he divided the light from the da
rkneses.
You typed the 89-character verse in 7.07162 seconds, which reflects a typing spe
ed of 151.026 WPM. Your error rate was 0.0224719 (not including backspaces) and
0.0449438 (including backspaces).
That concludes this multiplayer game. Here are the final WPM rankings:
Ken: 136.644
Kevin: 131.125
Keith: 123.1
Congratulations, Ken--you won!
Exported 12 test result(s) in 0.000694822 seconds.
Exported 189 word results in 0.00203796 seconds.
Exported 3 mean WPM rows in 9.1511e-05 seconds.
Type 'y' to call a Python script that will visualize your new multiplayer stats;
type 'n' to skip this process. (This may require some setup on your part; see R
eadme for more details. You can also run the multiplayer script separately if yo
u prefer.
```

Multi-computer multiplayer gameplay

If you have lots of people who wish to play a multiplayer game, then it may be more efficient to use a multi-computer approach. That way, lots of people can play the game at the same time, and they can all compete on the keyboard that they're most comfortable with. (However, given that each player will need to download and install TTTB on their computer beforehand, and that you'll need to perform some additional steps to compile everyone's results, this method may not necessarily save time.)

Here's how I recommend going about this method:

1. Have every player launch a *single-player game* within Type The Bible.
2. Decide the verse ID at which you would like to begin the test, then have each player select this ID within the 'I' single-player mode. (This mode, not to be confused with 'i' mode, begins at a specified verse, then automatically presents the tests that follow this verse—whether or not the player has typed them already. This helps ensure that each player will be typing the exact same set of tests—which will be crucial for your subsequent analyses to be valid.)
3. You'll also want to decide how many verses total to type. I recommend choosing a number that's a multiple of 10, since this will make the process of sharing results a bit easier. (You'll learn why shortly.)
4. Once players have finished typing their verses, they should all add their results *for only the races they just typed* to the same folder. This could be an online storage folder; a flash drive; or even a network-attached storage setup. (If your total test count is a multiple of 10, players can simply share their most recent autosave folder, which should contain only the races they just typed. Otherwise, they can simply cut and paste their results into a separate .csv file. Note that there's no need to include headers.)
5. A designated player should then copy all of these test results into his/her 'Files/MP_Test_Result_Files_To_Combine'

folder. Next, he/she should launch TTTB, then select 'c' within the game's start menu and follow the subsequent prompts. TTTB will then run some Python code that (1) converts the copies of multiplayer results into a single file and (2) runs the same visualizations script that it executes for single-computer multiplayer games.



```
kjb3@kjb3-lm: ~/D1V1/Documents/!Dell64docs/Programming/CPP/cpp_tttb/build
File Edit View Search Terminal Help
cd '/home/kjb3/D1V1/Documents/!Dell64docs/Programming/CPP/cpp_tttb/build'
(base) kjb3@kjb3-lm:~/D1V1/Documents/!Dell64docs/Programming/CPP/cpp_tttb/build$ ./tttb

Welcome to Version 1.0 of the C++ Edition of Type Through the Bible!
Please review the Readme for important gameplay instructions.

Type 's' or 'm' for single-player or multiplayer mode, respectively.
To import multiplayer results into your single-player files, type 'r.'
To combine multiplayer results within ../Files/MP_Test_Result_Files_To_Combine in
to a single file, then analyze them, type 'c'.
To exit the game, type 'e.'
Type the first and last verse IDs (inclusive) of the multiplayer session, separa
ted by an underscore, followed by Enter. (Example input: 30786_30795) Only resul
ts within this range of IDs will get incorporated into the results.

Alternatively, type 'y', followed by 'Enter', to have the game detect these IDs
within the first set of results that it analyzes.

y
Type 'y' to process these results and 'n' to cancel.
Calling Python script. (This can take a little while.)
Starting Python script. It may take a little while to run; please be patient.
Pandas version: 2.3.2   Numpy version: 2.3.2   Plotly version: 6.3.0
Combining multiplayer files together.
player_list: ['Ken3']
30786 and 30795 will be used as the starting and ending verse IDs, respectively.
These IDs are based on the verses typed by Ken3 within 20250725T130821_ken3test
_test_results.csv.
KJB3 within autosaved_test_results_Ken2_missing_verse.csv did not complete all o
f the tests between 30786 and 30795; thus, this player will be excluded from the
combined dataset.
Finished combining multiplayer files into a single file.
This file is available at ../Files/Multiplayer/20250725T130325_CMR_test_results.
csv.
Analyzing multiplayer data.
Analyzing results within 20250725T130325_CMR_test_results.csv.
Creating visualizations.
Finished calculating and visualizing multiplayer stats in 0.576 seconds.
```

Converting multiplayer data to single-player data

You can also use the 'r' command within the start menu to import multiplayer results into your single-player results if you so choose. (This will be particularly helpful if you completed a long stretch of races within a multiplayer game; importing them into your single-player file will then allow them to count towards your overall progress in typing the entire Bible.)

```
kjb3@kjb3-lm: ~/D1V1/Documents/Dell64docs/Programming/CPP/cpp_ttb/build
File Edit View Search Terminal Help
Finished updating configuration options.
Data for Ken will now be added to your main test results files under the name KJ
B3. Press 'y' to proceed or 'n' to cancel.
Exported 4 test result(s) in 0.000307687 seconds.
A revised copy of your multiplayer test result data was added to test_results.cs
v.
Exported 35839 Bible verses in 0.2536 seconds.
Your Bible verse file was also updated with the results of your multiplayer test
s.
Finished importing multiplayer word result data from:
../Files/Multiplayer/20250829T211930_mptest_test_results.csv
Exported 63 word results in 0.000222933 seconds.
A revised copy of your multiplayer word result data was added to word_results.cs
v. Multiplayer imports are now complete! However, visualizations of your single-
player results will not be updated until you next run TTB in single-player mode
.

Type 's' or 'm' for single-player or multiplayer mode, respectively.
To import multiplayer results into your single-player files, type 'r.'
To combine multiplayer results within ../Files/MP_Test_Result_Files_To_Combine in
to a single file, then analyze them, type 'c'.
To exit the game, type 'e.'
```

Analyzing your results

Type Through the Bible offers two main ways to review and evaluate your progress. The first are a series of .csv files that keep track of your results; the second are a set of interactive visualizations that make these results easier to interpret.

CSV files

Your single-player data is stored in the following .csv files within your game's Files folder:

1. 'test_results.csv' (This file provides detailed data on all of your test results, including WPM and accuracy data; start and end times; and verse-related information.)
2. 'CPDB_for_TTTB.csv' (A local copy of the Catholic Public Domain Bible. This file shows the full text of each verse, typed or untyped, as well as (1) the number of times you've typed it and (2) your highest-ever WPM for each verse.
3. 'word_results.csv' (This file provides WPM and accuracy data for each individual word that you typed within each test.)

Meanwhile, your multiplayer result data for each session can be found within Files/Multiplayer. These files will be prefaced with a timestamp and (if provided) a custom string in order to make it easier to link each file to each individual multiplayer session.

Test- and word-level results files are made available for each session, along with a simple '...pivot.csv') file that stores each player's mean WPM.

NOTE: Be very careful when opening these files in Excel! I've found that Excel tends to automatically reformat dates within spreadsheets, then save those reformatted dates. These reformatted versions will most likely raise errors when you attempt to use them again. (As an alternative, I recommend LibreOffice Calc, which doesn't appear to attempt to reformat dates. You could also try opening the .csv files within a text editor (since, in the end, they're just text files) in order to further prevent any reformatting nightmares.

HTML-based interactive charts

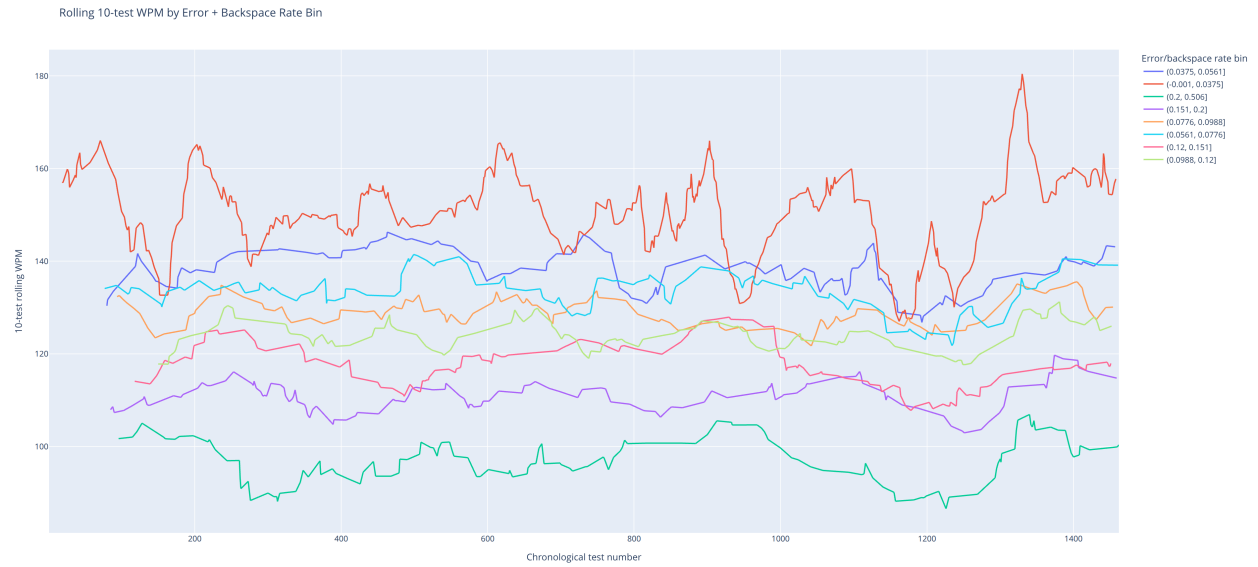
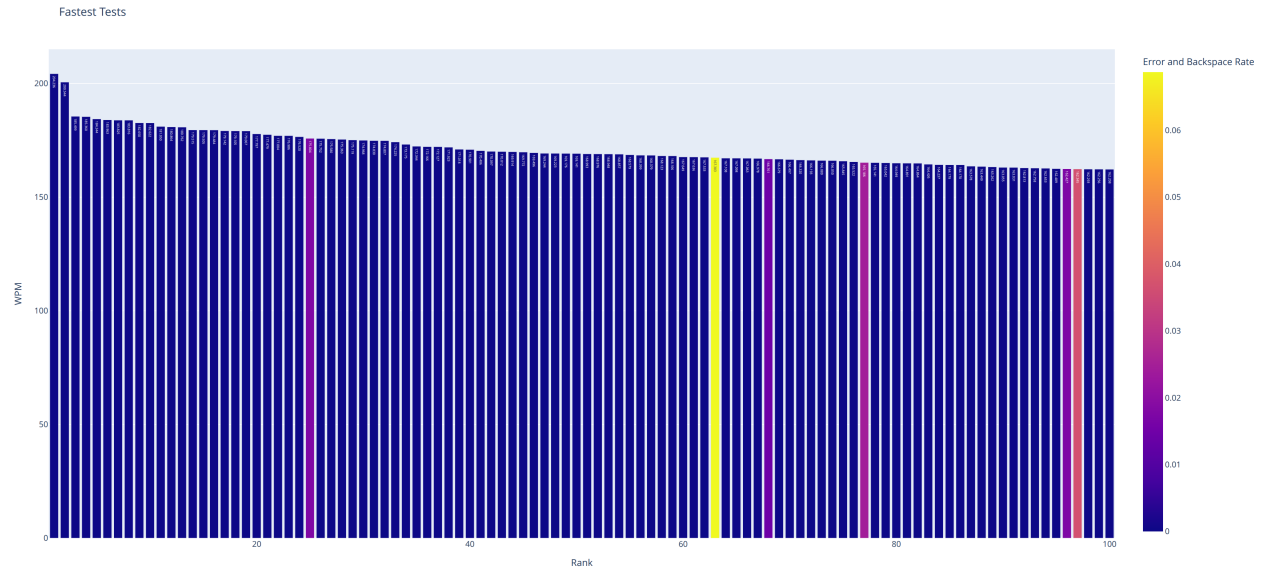
TTTB also provides many interactive, HTML-based visualizations of your results. These are created via a separate Python script that TTTB can call at the end of each single-player and multiplayer game. You can find these charts within the ‘Multiplayer’ and ‘Single_Player’ subfolders of the ‘Visualizations’ folder.

Because these visualizations are HTML-based, you can hover over bars and lines for more information; remove certain categories of lines/bars to focus on an area of interest; and also zoom in on particular parts of the graph.

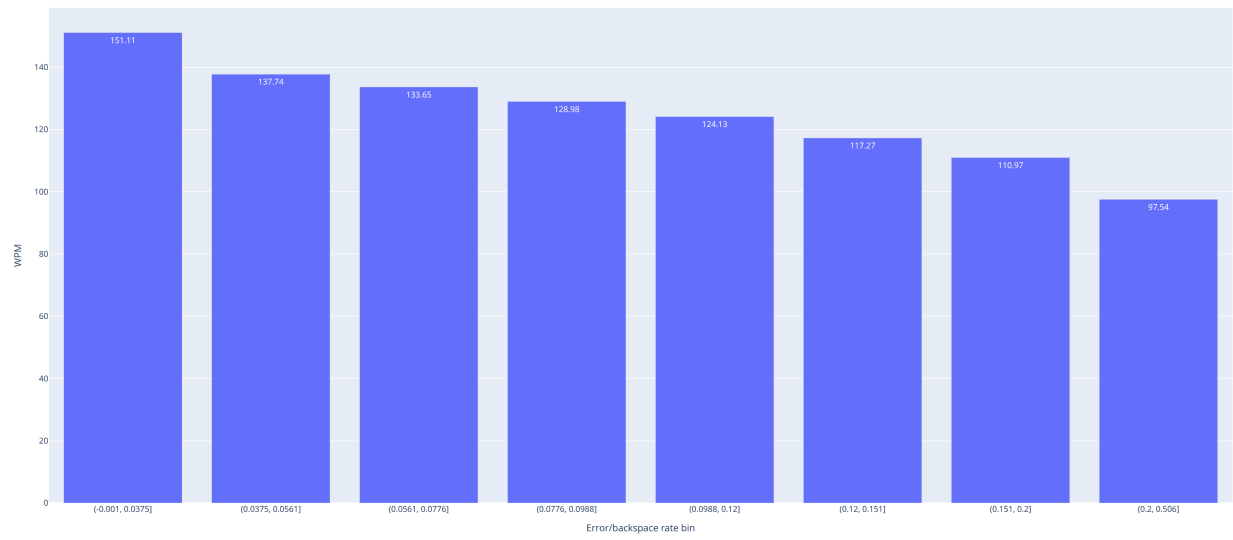
TTTB generates around 29 separate single-player visualizations and 4 multiplayer visualizations. They help you make sense of your average WPM; your accuracy; and your progress in typing through the whole Bible.

Here are some sample PNG screenshots: (Unlike the original HTML copies, these aren’t interactive.)

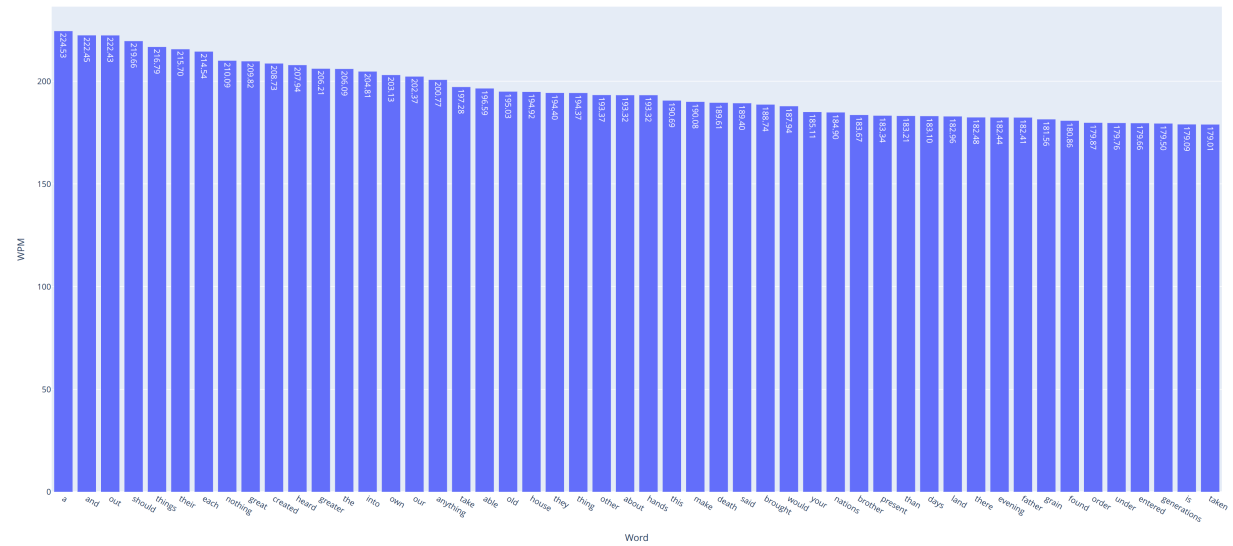


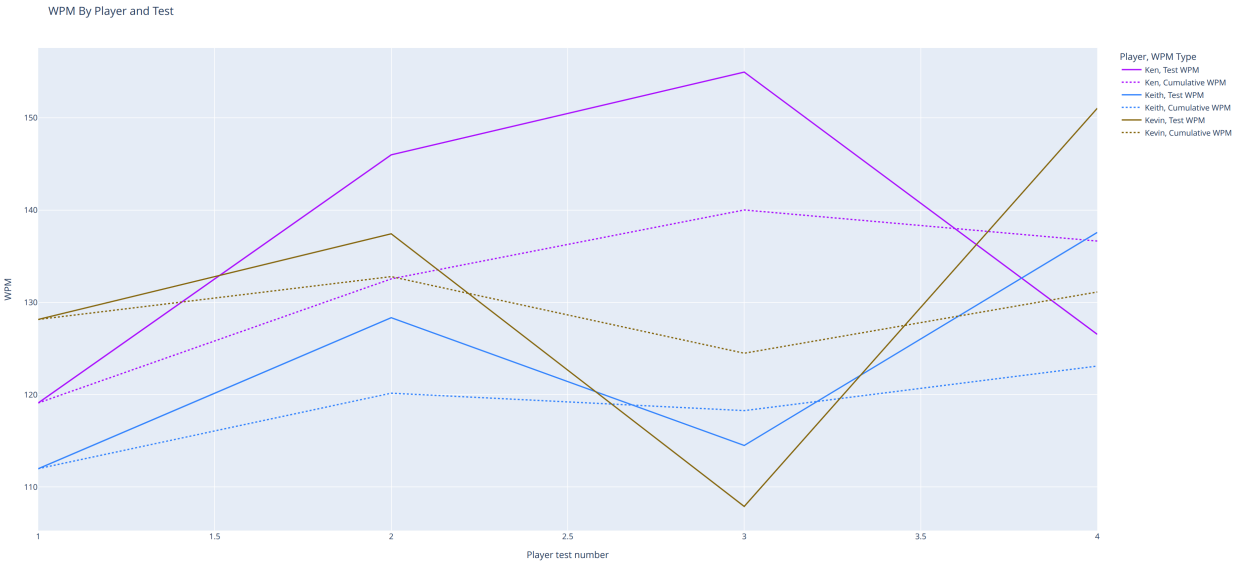
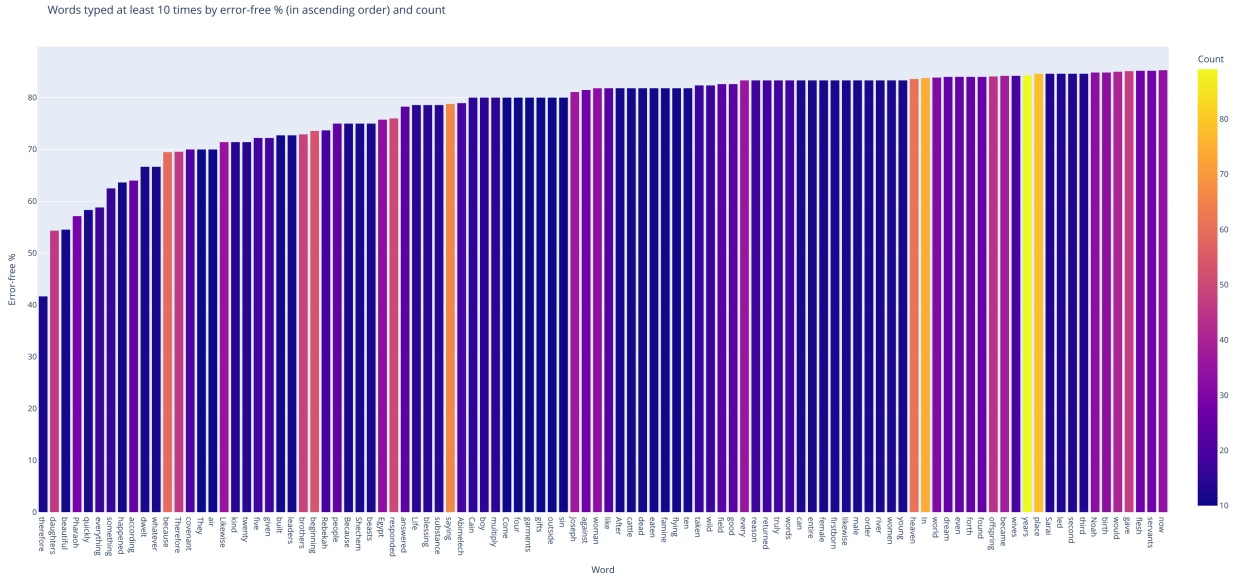


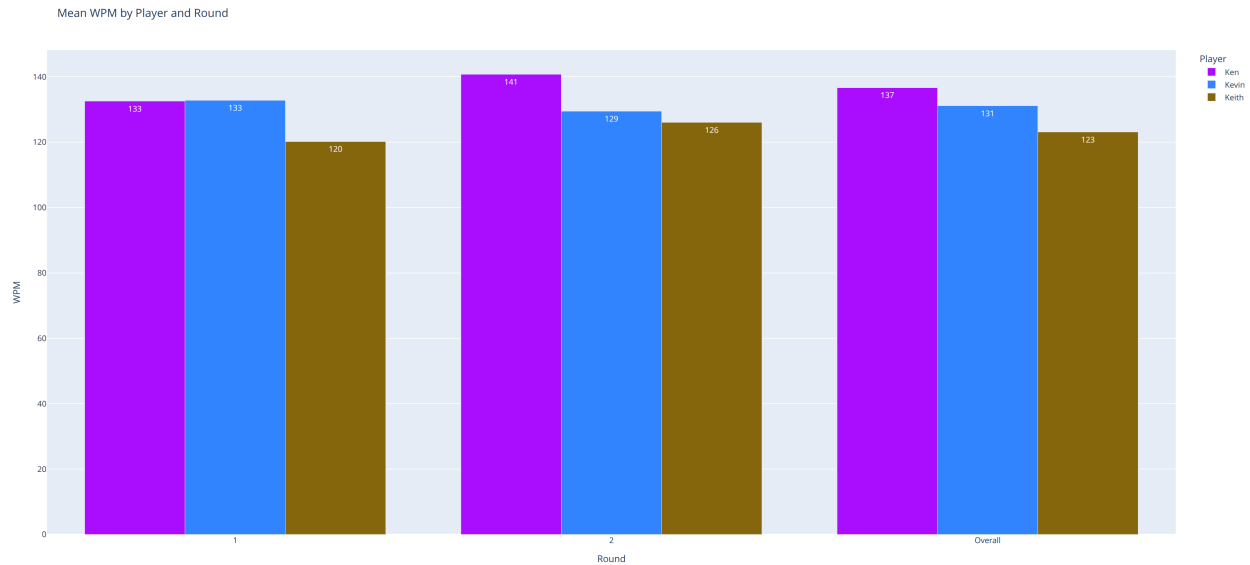
Mean WPM by Accuracy Bin



Highest word-level mean WPM for words typed at least 10 times







Reporting issues

I have completed over 1,400 tests on Type Through the Bible so far, along with quite a few multiplayer rounds. However, it's still possible (and perhaps, for a hobby project like this, likely) that bugs, typos, and other issues will appear.

Please make me aware of any issues, large or small, by posting them within https://github.com/kburchfiel/cpp_tttb/issues. If you don't have a GitHub account, you can also email me at kburchfiel3@gmail.com. (Please include 'TTTB' in your subject so that I know your message is related to this game.)

You are also welcome to use these channels to submit feature requests and improvements; however, since the time I have to work on this project is limited, I may not be able to implement your idea.

Development Notes

I had created an earlier Bible typing game in Python, but given my interest in learning how to apply C++ to create games, I figured that a C++ version of this game would be a great learning opportunity. Although I didn't reference the Python version of this game when writing this code, my experiences with that project certainly influenced the current one.

This project was a lot of fun to work on. While it doesn't feature a GUI, it *was* a good opportunity to for me to get reacquainted with C++—and to gain experience with importing and exporting CSV data. I hope that you enjoy playing it as much as I did coding it!

As with my other GitHub projects, I chose not to use generative AI tools when creating Type Through the Bible. I wanted to learn how to perform these tasks in C++ (or in a C++ library) rather than simply learn how to get an AI tool to create them.

This code also makes extensive use of the following open-source libraries:

1. Vincent La's CSV parser (<https://github.com/vincentlaucsb/csv-parser>)
2. CPP-Terminal (<https://github.com/jupyter-xeus/cpp-terminal>)

In addition, this program uses the Catholic Public Domain Version of the Bible that Ronald L. Conte put together. This Bible can be found at <https://sacredbible.org/catholic/>. I last updated my local copy of this Bible (whose text does get updated periodically) around June 10, 2025.

Acknowledgments

I'd like to give special thanks to Ronald L. Conte Jr. for his work on the [Catholic Public Domain Version](#) of the Bible (the translation that TTTB uses).

I am also grateful to Vincent La for [his excellent C++ csv parser](#) and to the team behind the [cpp-terminal](#) library. These two libraries play a major role in C++ TTTB. I'm also grateful to the Plotly team for their excellent [Python visualization library](#).

Finally, this game is dedicated to my wife, Allie. I am very grateful for her patience and understanding as I worked to put it together! My multiplayer gameplay sessions with her (yes, she was kind enough to play it with me) also helped me refine the code and improve the OSX release.

Blessed Carlo Acutis, pray for us!

Appendix: Compilation Instructions

If you'd like to modify or build upon this game, you'll probably need to compile it first. (Note that there are two separate scripts that you'll need to compile: `tttb.cpp`, the game's main C++ file, and `tttb_py_installer.py`, a Python file.)

The following compilation instructions apply to Linux, OSX, and Windows, though the steps will differ slightly across platforms. Please let me know if you encounter any issues.

1. Download [CMake](#) if you haven't already.
2. Clone the [cpp-terminal](#) and [csv-parser](#) libraries, both of which are permissively licensed, and build them using CMake. **Note:** I recommend switching the `BUILD_SHARED_LIBRARIES` option within the `cpp-terminal CMakeLists.txt` to `OFF` for all platforms in order to make your resulting code more portable.

(To build each library using CMake, (a) create a 'build' folder within the folder on your computer that contains the library; (b) navigate within this folder using your terminal; (c) run `cmake ..` in preparation for the build; and then, if that command was successful, (d) run `cmake --build .` Don't forget the space and period at the end!)

3. Copy these folders to a directory of your choice, or leave them in your downloads folder if you prefer.
4. Clone [Type Through the Bible](#) and move it to a directory of your choice.
5. **IMPORTANT:** Within TTTB's `CMakeLists.txt` file, you *must* replace the existing paths to my copies of the `cpp-terminal` and `csv-parser` libraries within the `add_subdirectory()` calls with paths to *your* copies of these folders (at least for the platform(s) for which you're building TTTB). Within each `add_subdirectory()` call, the first path is to the actual 3rd-party-library's location on your computer; the second path tells CMake where to place some additional files related to that library. (You don't need to create this second folder on your system; CMake will create it automatically.)
6. Navigate into TTTB's `build/` folder.
7. Use CMake to build TTTB. (See above steps for reference if needed.) Once it has been build, copy the resulting executable (e.g. `tttb` or `tttb.exe` on Windows) into your build folder. (Windows will likely place it into a separate subfolder, but it must be placed within your main build folder in order for the relative paths used by the program to work correctly.)
8. Next, you'll need to build an executable version of the `tttb_py_complement.py` file; this can be accomplished via Pyinstaller. First, you'll want to download [Pyinstaller](#) if you don't have it already. Also, download Python (i.e. via [Miniforge](#) if it's not already on your system).
9. Make sure that **Pandas**, **Numpy**, and **Plotly** are all installed within the Python environment that you plan to use to run Pyinstaller. It's helpful, but not necessary (to the best of my knowledge), to have them within your base environment. (These can be downloaded via `conda-forge`.) Otherwise, the executable version of your `tttb_py_complement.py` file won't work correctly.

Note: You may also want to install JupyterLab (e.g. via `conda install jupyterlab`) so that you can modify, if needed, the .ipynb notebooks contained within the source code. (Just make sure to export these updated notebooks to a .py file so that they can get processed correctly by `create_release_folder.py` and your `pyinstaller` call.)

10. Once you have Python and Pyinstaller set up, first activate your environment of choice within your terminal (e.g. by running `conda activate tttb_env`; replace 'tttb_env' with whatever your environment name happens to be.) Next, navigate to TTTB's 'build' folder and run `pyinstaller tttb_py_complement.py`. This will create an executable version of this file, along with an '_internal' folder that contains important library components, into a 'dist/tttb_py_complement' subfolder within your 'build' folder. ****Cut and paste both the _internal folder and the tttb_py_complement executable into your build folder, as that's where the program will look for them.****

Note: If you encounter issues with Pyinstaller, consider (1) creating a brand new conda environment that contains *only* the latest copies of the libraries required for the code to run (i.e. Pyinstaller, Numpy, Pandas, and Plotly) and their dependents, then (2) rerunning your Pyinstaller command from within that conda environment. (I've found, based on my tests, that Pyinstaller will use the versions of the libraries present within the conda environment from which you activate it. Therefore, if your base environment is giving you trouble, simply create a new environment instead.)

Here's a sample set of terminal commands that can help you accomplish these steps:

[Conda will need to be added to your path in order for the following step to work; this was probably already completed when you installed Miniforge.]

a. `conda activate base`

b. `conda create -n tttb_analysis python pandas numpy plotly pyinstaller`

[After running the above code, make sure to enter 'y' after Conda asks whether you wish to download and install these packages (along with their dependencies).]

c. `conda activate tttb_analysis`

[Navigating to the location of the Git clone's build folder: (Replace this path with the path to your own Type Through the Bible build folder).]

d. `cd '/home/kjb3/D1V1/Documents/!Dell64docs/Programming/CPP/cpp_tttb/build'`

[Running pyinstaller:]

e. `pyinstaller tttb_py_complement.py`

10. Navigate up to the main `cpp_tttb` folder (e.g. via `cd ..` in your terminal, assuming you're still in the build folder) and run `python create_release_folder.py neither`. This will create a new copy of TTTB with blank output files rather than the existing files within the TTTB directory.

Note: The `create_release_folder.py` file also contains code that can help automate the process of building and (where necessary) moving your C++ and Python executables. (For instance, you can pass **both** as a corresponding argument to the file rather than **neither**, the default, in order to compile the executables. See the source code for more documentation and details.) However, I strongly recommend performing these steps outside of this file the first time around so that you can more easily identify and debug any issues. Also note that, if you set 'both' or 'py' as your argument, you'll want to call this script within the Python environment that you want to use for Pyinstaller.)

If you do want to try compiling both the C++ and Python executables via the 'create_release_folder.py' script, first complete steps a, b, and c in the above list. Next:

- d. Navigate to TTTB's root folder via the following command: (Make sure to replace the path shown with the path to your own copy of TTTB. Note that the root folder won't necessarily be named 'cpp_tttb'.)

`cd '/home/kjb3/D1V1/Documents/!Dell64docs/Programming/CPP/cpp_tttb'`

e. Execute the following command:

```
python create_release_folder.py both
```

11. Try launching the TTTB executable—if it works, congratulations! If not, double-check the steps and/or reach out to me with any questions.